Lissajous Ground-Scan Implementation

SimGroundLissajous

Purpose

The `SimGroundLissajous` operator simulates a smooth ground-based Lissajous boresight scan for a TOAST observation. It follows the same broad workflow as a ground-scan operator: it reads a ground schedule, creates observations, generates timestamps and boresight pointing, converts the mount-frame pointing to sky coordinates, and records useful observation metadata.

The important design choice is that the scan is generated natively in the mount coordinate system: azimuth and elevation. The equatorial pointing is then derived afterward by converting the generated Az/El boresight at each timestamp.

Coordinate Picture

The scan is first described in horizontal coordinates:

- Azimuth, Az, is the commanded horizontal mount angle.
- Elevation, El, is the commanded elevation mount angle.
- RA/Dec are produced later by converting the time-dependent Az/El boresight through the observing site and timestamp.

So the operator is not scanning directly in equatorial coordinates. It scans in Az/El and then asks: "where on the sky was this mount pointing at this time?"

Schedule Rows and Observations

Each schedule row supplies a time interval and a nominal mount pointing range:

Schedule quantity	Role in the Lissajous scan
Start / stop time	Defines the time span of one generated scan.
$\mathrm{Az}_{\mathrm{min}}$, $\mathrm{Az}_{\mathrm{max}}$	Define the default azimuth center and, optionally, the default azimuth amplitude.
El	Defines the default elevation center.
Rotation / boresight angle	Applied as a boresight rotation after the Az/El pointing is generated.

For each schedule row, the operator creates one Lissajous scan over that row's time interval. With the default zero drift rates, the scan center is fixed in mount coordinates during that observation. The sky moves through that fixed mount pattern as Earth rotates.

At the next schedule row, the operator recomputes the center from the new scheduled $\mathrm{Az}_{\min}$, $\mathrm{Az}_{\max}$, and El, then starts a new Lissajous pattern.

Core Scan Equations

For each observation, the operator defines a local elapsed time

$$t = \frac{i}{f_{\mathrm{samp}}},$$

where i is the sample index and f_{samp} is the detector sample rate.

The azimuth and elevation centers are

$$\mathrm{Az}_0 = \frac{\mathrm{Az}_{\min} + \mathrm{Az}_{\max}}{2}, \qquad \mathrm{El}_0 = \mathrm{El}_{\mathrm{schedule}}.$$

The Lissajous trajectory is then

$$\mathrm{Az}(t) = \mathrm{Az}_0 + v_{\mathrm{Az,drift}} t + A_{\mathrm{Az}} \sin\left(2\pi f_{\mathrm{Az}} t + \phi_{\mathrm{Az}}\right),$$

$$\mathrm{El}(t) = \mathrm{El}_0 + v_{\mathrm{El,drift}} t + A_{\mathrm{El}} \sin\left(2\pi f_{\mathrm{El}} t + \phi_{\mathrm{El}}\right).$$

Here A is amplitude, f is oscillation frequency, ϕ is phase, and v_{drift} is an optional linear drift of the scan center.

What Zero Drift Means

By default,

$$v_{\mathrm{Az,drift}} = 0, \qquad v_{\mathrm{El,drift}} = 0.$$

This means the Lissajous pattern is fixed around the scheduled Az/El center for the duration of one observation. The telescope mount repeats the same mount-frame pattern, while the sky drifts through it.

In plain language:

> Within one observation, hold the scheduled mount center and draw a Lissajous around it.
> At the next schedule row, use the newly scheduled center and draw a new Lissajous pattern.

This is discrete re-centering between observations, not continuous source tracking inside a single observation.

What Nonzero Drift Does

The drift terms move the center of the Lissajous pattern linearly during an observation. For example, a positive elevation drift raises the entire pattern in elevation as time passes.

These drift rates can be used to approximate tracking, to deliberately sweep a larger region, or to move the pattern from one scheduled center toward another. They do not automatically compute the source tracking rate, and they do not guarantee that the same RA/Dec path is retraced.

A nonzero drift usually makes the mount path an "open" Lissajous sweep rather than a closed repeating figure centered at one fixed Az/El position.

Default Amplitudes and Centers

The azimuth amplitude has a schedule-aware default:

$$A_{\mathrm{Az}} = \frac{|\mathrm{Az}_{\max} - \mathrm{Az}_{\min}|}{2} \quad \text{if} \quad \texttt{lissajous_az_amplitude = None}.$$

If `lissajous_az_amplitude` is explicitly set, that value is used instead.

The elevation amplitude defaults to the configured trait value. It is not derived from the schedule because the schedule provides a single elevation center rather than an elevation min/max range.

Mount Motion Limit Checks

The operator checks the requested Lissajous motion against user supplied mount motion limits. These limits are mechanical mount-axis limits, not on-sky scan speed limits.

The peak azimuth and elevation mount rates are computed analytically:

$$\left|\dot{\mathrm{Az}}\right|_{\max} = |v_{\mathrm{Az,drift}}| + 2\pi f_{\mathrm{Az}} \, |A_{\mathrm{Az}}| \, ,$$

$$\left|\dot{\mathrm{El}}\right|_{\max} = |v_{\mathrm{El,drift}}| + 2\pi f_{\mathrm{El}} \, |A_{\mathrm{El}}| \, .$$

The peak accelerations are

$$\left|\ddot{\mathrm{Az}}\right|_{\max} = (2\pi f_{\mathrm{Az}})^2 \, |A_{\mathrm{Az}}| \, ,$$

$$\left|\ddot{\mathrm{El}}\right|_{\max} = (2\pi f_{\mathrm{El}})^2 \, |A_{\mathrm{El}}| \, .$$

These peak values are compared to `max_mount_rate_az`, `max_mount_rate_el`, `max_mount_accel_az`, and `max_mount_accel_el`.

Because the scan is generated in Az/El, this is a mount-coordinate check:

$$\dot{\mathrm{Az}} \quad \text{and} \quad \dot{\mathrm{El}}.$$

It is not an on-sky angular speed check. A small-angle on-sky speed would look approximately like

$$v_{\mathrm{sky}} \simeq \sqrt{\left(\cos(\mathrm{El})\,\dot{\mathrm{Az}}\right)^2 + \dot{\mathrm{El}}^2}.$$

That cos(El) projection is intentionally not applied in the mount limit check.

Main User-Facing Traits

Trait	Meaning
`lissajous_az_amplitude`	Azimuth oscillation amplitude. If `None`, use half the scheduled azimuth throw.
`lissajous_el_amplitude`	Elevation oscillation amplitude.
`lissajous_az_rate`	Azimuth oscillation frequency.
`lissajous_el_rate`	Elevation oscillation frequency.
`lissajous_az_phase`	Azimuth sinusoid phase offset.
`lissajous_el_phase`	Elevation sinusoid phase offset.
`lissajous_az_drift_rate`	Linear drift rate of the azimuth scan center.
`lissajous_el_drift_rate`	Linear drift rate of the elevation scan center.
`check_mount_motion_limits`	Enable or disable mount rate and acceleration checks.
`max_mount_rate_az`	Maximum allowed azimuth mount rate.
`max_mount_rate_el`	Maximum allowed elevation mount rate.
`max_mount_accel_az`	Maximum allowed azimuth mount acceleration.
`max_mount_accel_el`	Maximum allowed elevation mount acceleration.

Operational Traits

Trait	Meaning
telescope	Telescope object with focalplane and site information.
telescope_file	Optional instrument file used when a telescope object is not supplied.
schedule	GroundSchedule object used to define observation rows.
schedule_file	Optional schedule file used when a schedule object is not supplied.
sort_schedule_file	Sort a loaded schedule by mean boresight RA.
weather	Built-in weather site name or weather HDF5 file.
realization	Weather realization index.
max_pwv	Optional maximum PWV for simulated weather.
median_weather	Use median weather parameters instead of sampling distributions.
use_ephem	Use PyEphem for Az/El to RA/Dec conversion.
use_qpoint	Use qpoint for Az/El to RA/Dec conversion.
distribute_time	Distribute observations along time rather than detector axis.
sample_set_size	Samples per sample-set when distributing in time.
detset_key	Focalplane column used to group detectors into detector sets.

Produced Data

For each observation, the operator writes shared data columns for:

- timestamps,
- azimuth,
- elevation,
- boresight quaternion in Az/El,
- boresight quaternion in RA/Dec,
- observatory position,
- observatory velocity,
- optional shared flags.

It also records scan metadata such as the realized minimum and maximum Az/El values and marks the scan type as `lissajous`.

What This Implementation Does Not Do

- It does not continuously solve for source tracking inside each observation.
- It does not automatically compute drift rates from a target's apparent Az/El motion.
- It does not check absolute mechanical pointing bounds such as $\mathrm{Az}_{\mathrm{min}}^{\mathrm{mount}}$ or $\mathrm{El}_{\mathrm{min}}^{\mathrm{mount}}$.
- It does not check total on-sky angular speed.
- It does not generate physical turnaround periods. For compatibility with TOAST operators that expect the standard key, it creates an empty `turnaround` interval list.

Practical Mental Model

The simplest way to understand the current implementation is:

> The schedule tells the telescope where to place the center of each pass in Az/El. The Lissajous traits tell it how to wiggle around that center. With zero drift, the sky passes through that mount-fixed pattern. At the next schedule row, the pattern is re-centered and the process repeats.

This makes the operator useful for simulating compact, smooth, repeated mount-frame patterns while still producing the RA/Dec pointing needed by later mapmaking and sky-projection steps.